

DEREK F. DILEO

Android Software Engineer | Kotlin · Jetpack Compose · UI Architecture

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SUMMARY

Android Software Engineer and UI architecture owner specializing in Kotlin, Jetpack Compose, MVI, and Clean Architecture. Proven record of modernizing production Android systems, establishing scalable UI and testing practices, and architecting complex hardware-integrated workflows. Experienced across Android, JNI/C++, persistence, automated testing, and full-stack development.

EXPERIENCE

TOMAHAWK ROBOTICS (acquired by AEROVIRONMENT in 2023)

Software Engineer II, Applications | [07/2025] – Present

Software Engineer I, Applications | [10/2023] – [07/2025]

Junior Software Engineer, Applications | 08/2022 – [10/2023]

Kotlin, Java, Jetpack Compose, Material 3, MVI, Clean Architecture, Hilt, Room, Coroutines/Flow, JNI/C++, JUnit, React, TypeScript, AWS

- **Serve as the Android UI architecture owner, leading modernization of the Android codebase and establishing development patterns across the team;** drove the transition from imperative Java, LiveData, and observer-based patterns to Jetpack Compose, Material 3, MVI, Clean Architecture, Hilt, reusable UI components, and layered architecture.
- **Expanded automated testing across the Android stack**, introducing pure Kotlin unit, instrumented, and Jetpack Compose UI testing with JUnit and AssertK to improve maintainability and regression confidence.
- **Architected and delivered turnkey vehicle-pairing applications that replaced manual CLI configuration with automated UI workflows**, dynamically discovering and configuring ground radios as operators enable or switch between vehicle nodes.
- **Integrated JNI to bridge native C++ libraries with Android components**, enabling Android applications to leverage existing native capabilities.
- **Built and deployed a React/TypeScript customer self-service portal on AWS GovCloud**, replacing manual hardware registration and software licensing with secure 24/7 activation and pre-provisioned entitlements, significantly reducing support overhead.
- **Integrated Jira APIs to automate customer bug reporting and issue creation**, eliminating manual ticket-entry workflows.

RX DELIVERED NOW

Quality Assurance Internship | 03/2022 – 08/2022

Flutter/Dart, Firebase, Google Analytics, Jira, Confluence

- **Led QA across mobile and web applications**, partnering with engineering and product teams to identify and drive defects through resolution while implementing analytics and conversion-tracking tooling.

VALENCIA COLLEGE – SCHOOL OF ARTS & ENTERTAINMENT | Orlando, FL

Lab Assistant II | Systems & Dante Network Administration | 09/2015 – 08/2021

Linux, Docker, VirtualBox, Dante Domain Manager, Cisco Networking, macOS, Windows

- **Expanded the role beyond traditional lab support by proactively improving campus production infrastructure**, taking ownership of systems administration, networking, technical documentation, and support.
- **Designed and deployed what became the largest educational RedNet network in North America**, using Linux virtualization, Dockerized infrastructure, and proprietary networking tools to support isolated virtual audio environments.
- **Administered production infrastructure** spanning Linux, macOS, Windows, Cisco networking, professional audio, and KVM systems while supervising student lab assistants.

SELECTED CAPSTONE

CITY OF ORLANDO – UNHOUSED YOUTH SOLUTIONS | 04/2022 – 08/2022

Technical Lead & Full-Stack Developer

Java, Spring Boot, Spring Security, JPA/Hibernate, MySQL, AWS RDS

- **Led a development team in designing and delivering a full-stack application for the City of Orlando**, owning technical coordination, task delegation, code reviews, and direct stakeholder requirements.
- **Designed the application's relational data model and persistence layer** while translating municipal requirements into features supporting identification and outreach efforts for unhoused youth.

TECHNICAL SKILLS

Android: Kotlin, Java, Jetpack Compose, Material 3, Android Views/XML, MVI, MVVM, Clean Architecture, Coroutines/Flow, Room, Hilt, JNI

Testing & Development: JUnit, AssertK, Compose UI Testing, TDD, Gradle, Git, GitHub/GitLab, Jenkins, Jira

Backend & Cloud: AWS/GovCloud, SQL, Spring Boot, REST APIs, Hibernate/JPA

Web & Cross-Platform: TypeScript, JavaScript, React, Flutter/Dart, Node.js

Infrastructure & Tools: Docker, Linux, Postman, Claude Code, ChatGPT

EDUCATION

VALENCIA COLLEGE | Orlando, FL

B.A.S., Computer Engineering & Software Development | 2022 — Graduated with Honors

A.S., Audio Engineering | 2020

A.S., Sound Production | 2020